



When A Face Feels Familiar: How Quality of Social Connections Influence Adolescent Brain Responses to Friend's Faces

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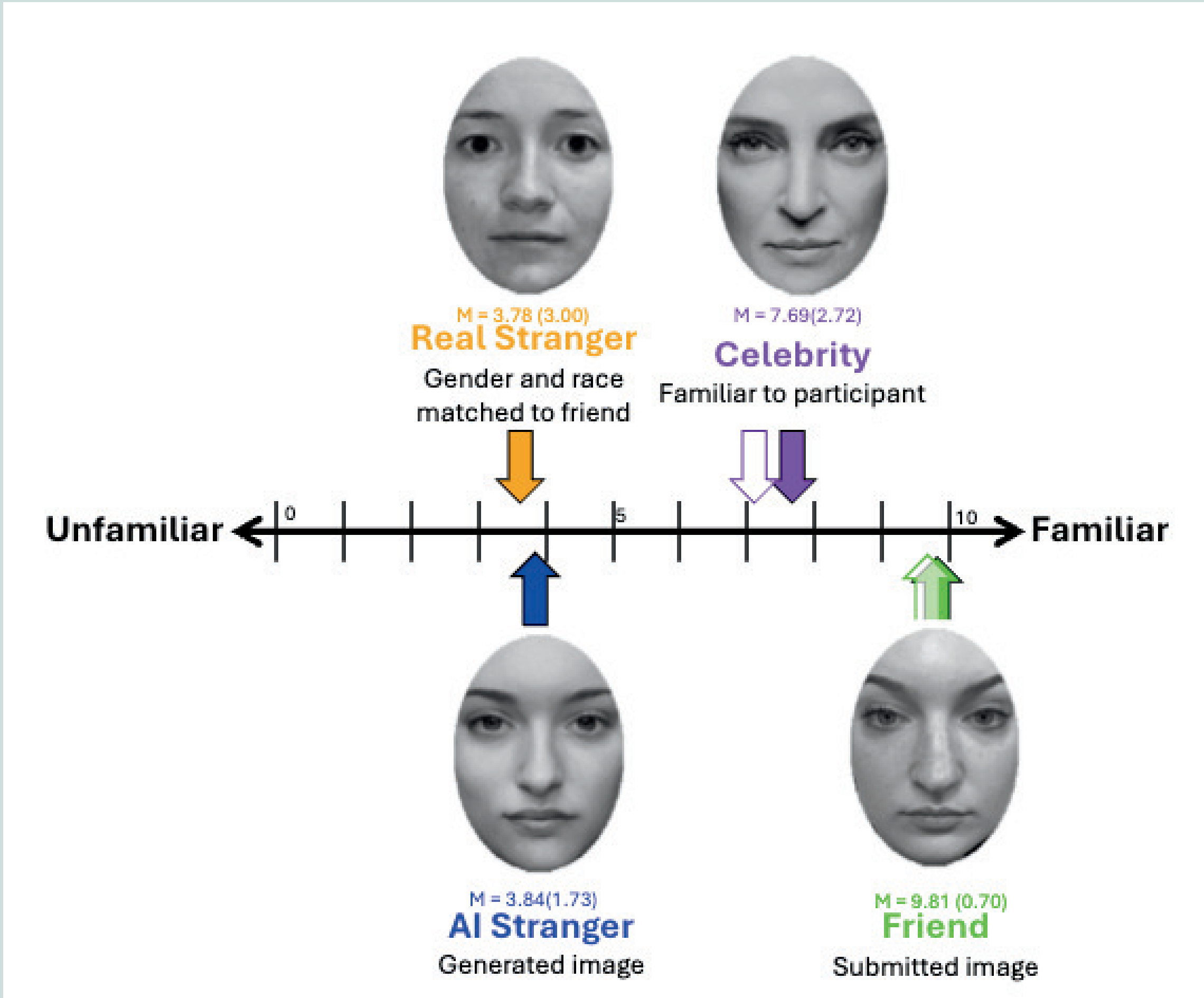


BACKGROUND

Adolescence is a critical period of social development marked by heightened sensitivity to social stimuli, including faces (Blakemore et al., 2008). Faces provide a wealth of information related to trust, safety, and inclusion, and the brain has specific mechanisms for recognizing familiar faces. Strong peer relationships during adolescence are critical for emotional well-being, self-esteem, and overall social development (Prinstein & Giletta, 2020), thus, how the brain differentiates between friends, other familiar faces, and strangers may explain how adolescents develop friendships. Additionally, adolescents are spending more time interacting with artificial intelligence (AI) generated images (Chong et al., 2024), and it may be critical to know how brain mechanisms differentiate between real and AI-generated faces. Our goals were to (1) evaluate brain responses to varying levels of facial familiarity and AI-generated images and (2) explore how trait-level differences in friendship quality influence these responses.

OBJECTIVES

- 1 Characterize neural responses to varying levels of familiarity
- 2 Assess how these responses change as a function of social support and relationship quality

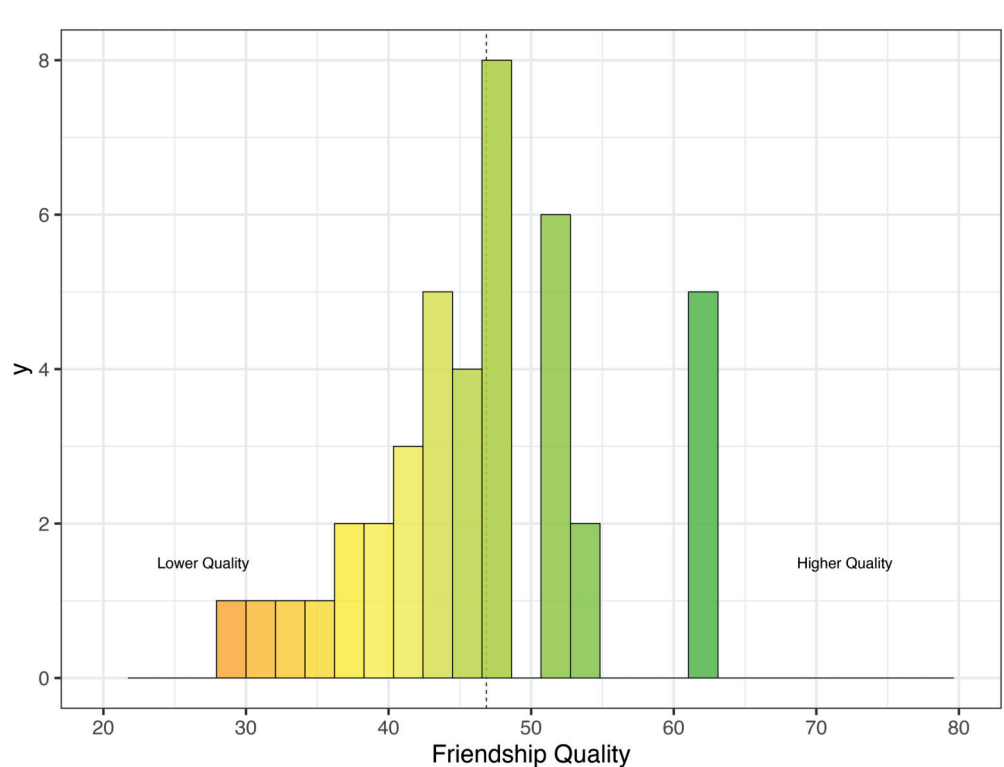


METHODS

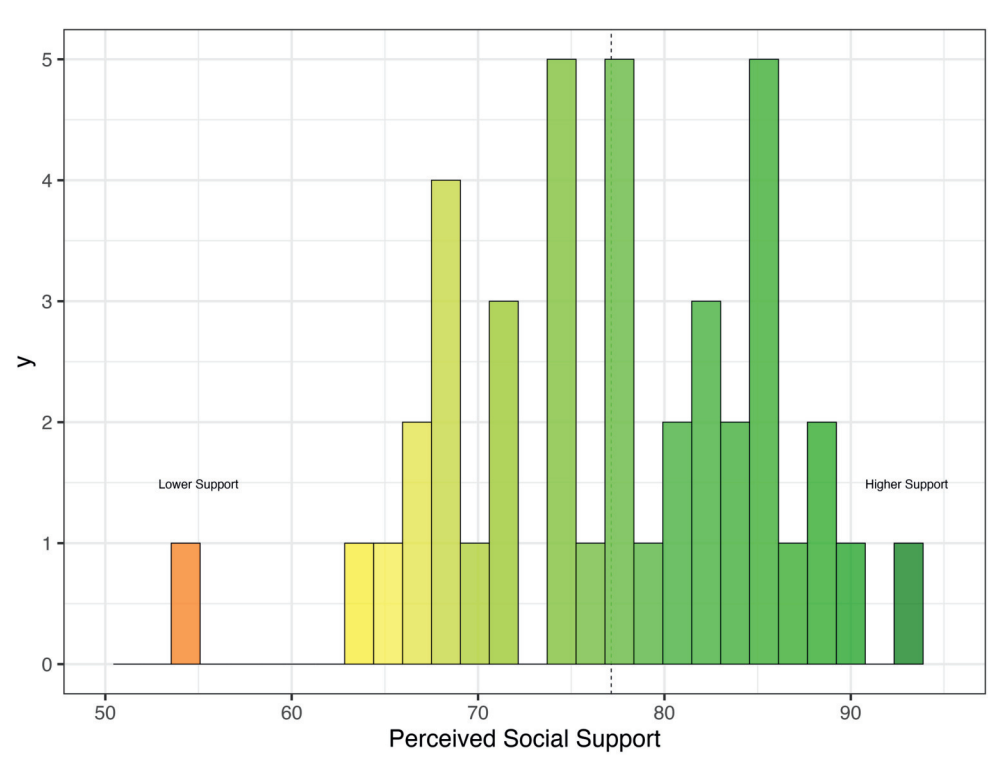
Adolescents N=51, 12-17 years (M = 14.2, SD= 1.6)

- 64% white, 18% black, 10% mixed-race
- 14% Hispanic/Latine
- 51% Female, 49% Male

- **P3 ERP Component** = attention allocation, stimulus evaluation, and familiarity processing.
 - Amplitude = magnitude of neural engagement
 - Latency = Speed of stimulus evaluation and categorization (285ms - 500ms)
- Two survey measures:



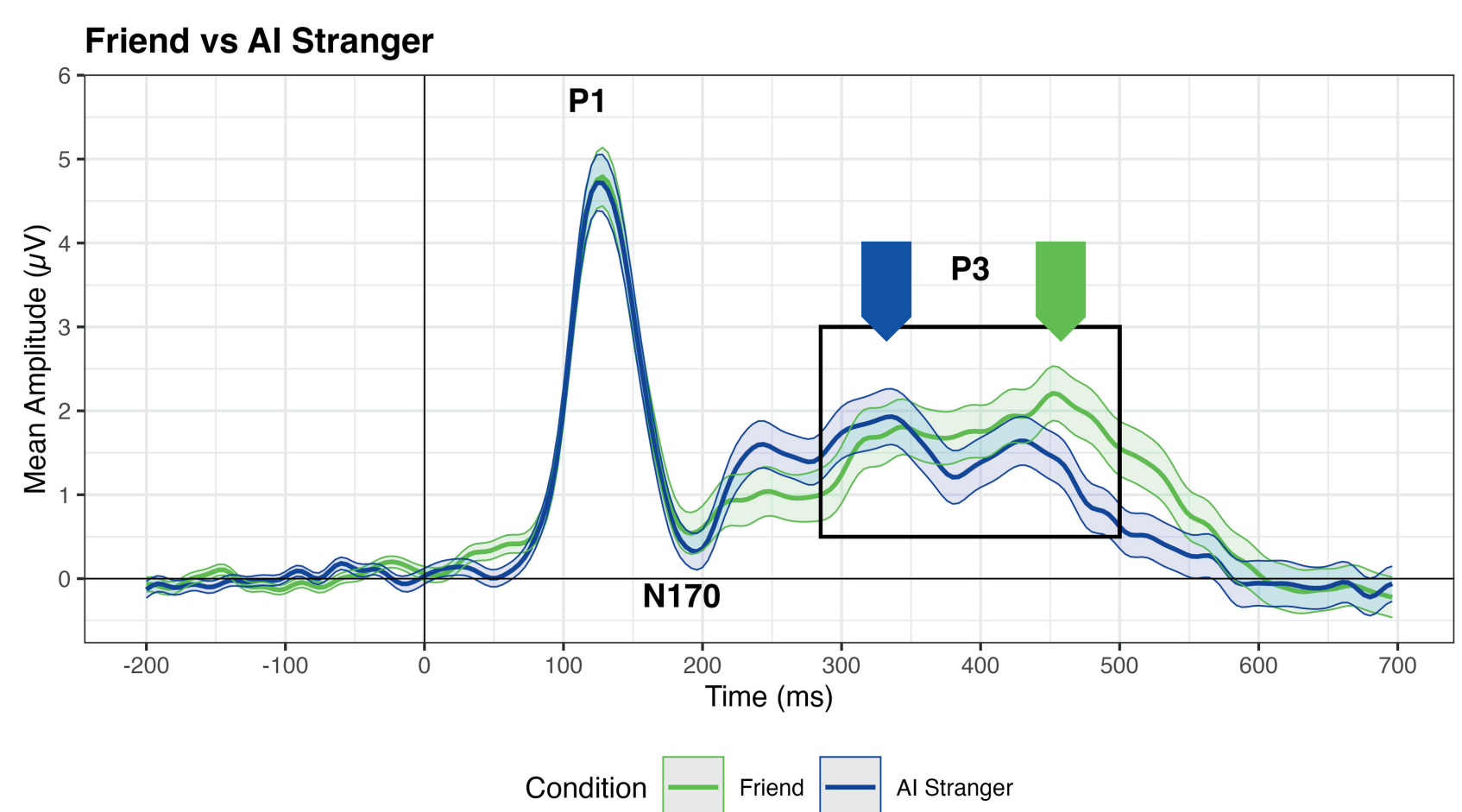
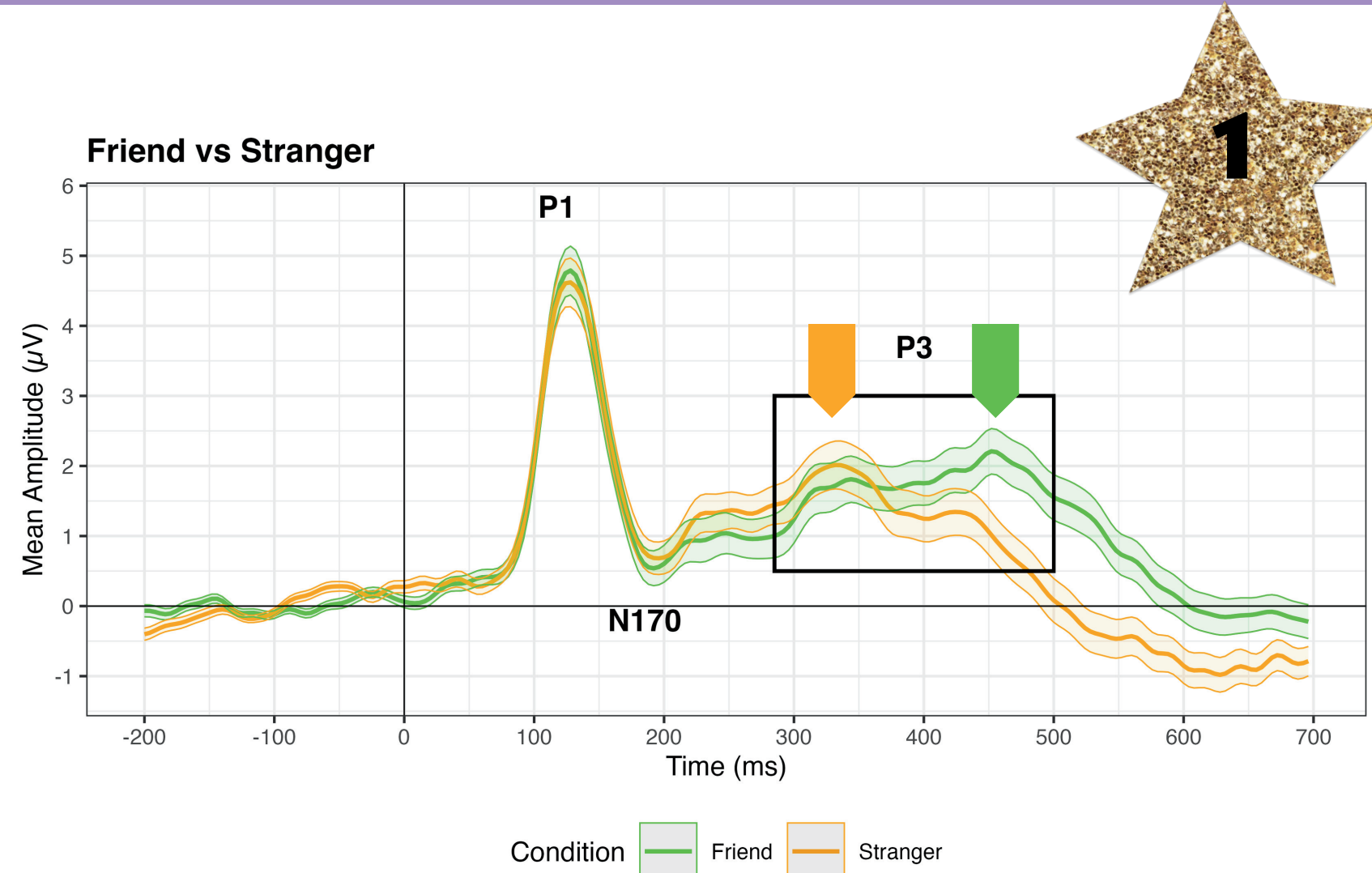
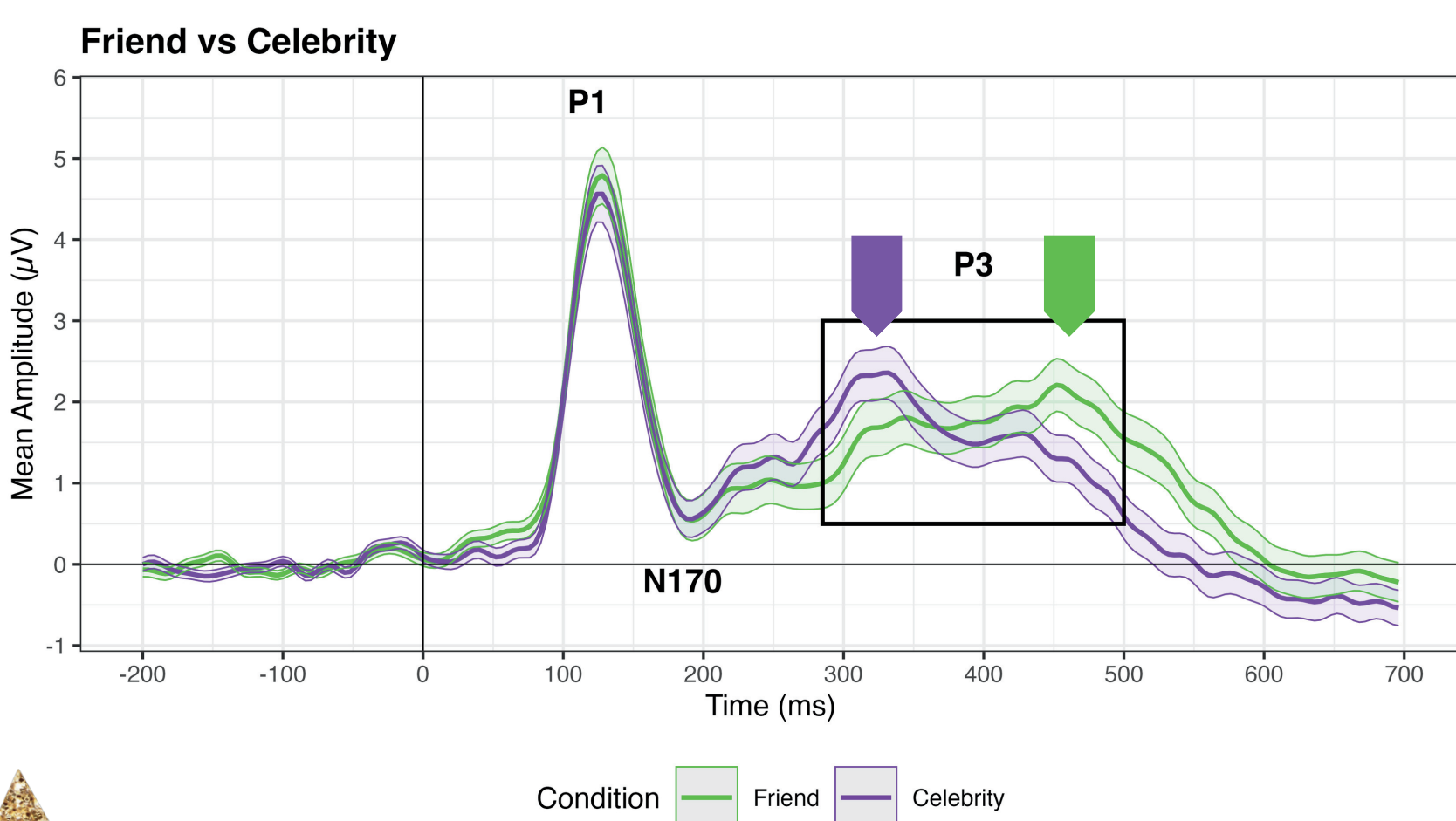
Friendship quality: NIH PROMIS Friend (Irwin et al., 2013)



Perceived social support: Social Provision Scale (Cutrona & Russell, 1987)

RESULTS

While the magnitude of neural engagement was comparable across conditions, the timing of neural responses differed. **Friend faces elicited significantly delayed responses relative to all other face types** ($p < .0001$, P3 latency), suggesting prolonged processing of personally meaningful social stimuli.

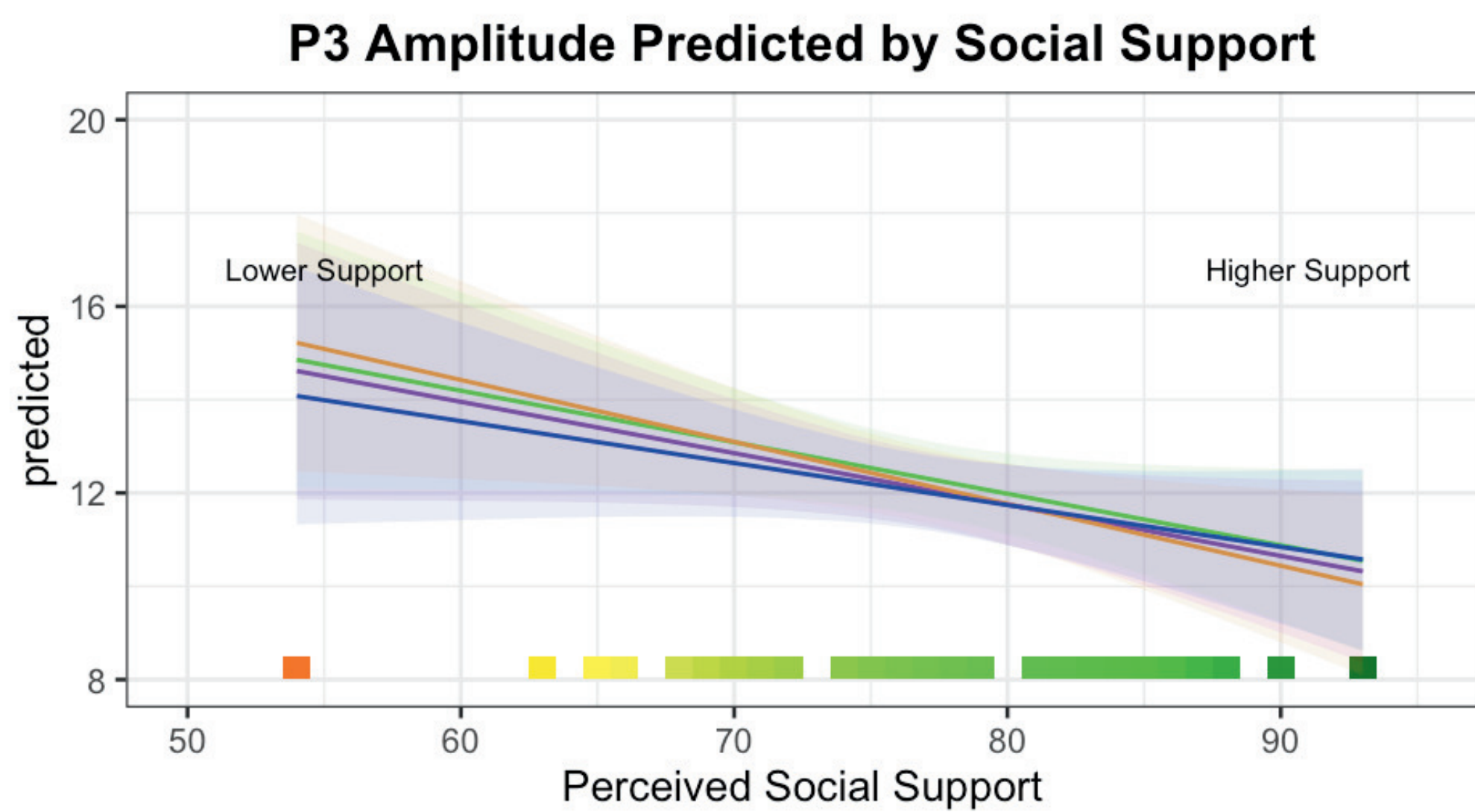


Increased social support was linked to decreased cognitive processing (P3 amplitude): $F(1, 38) = 7.44$, $p = .0096$:

- ↑ Social support ↓ P3 amplitude

The effect was significant for 3 out of 4 conditions, $ps < .049$

- ↑ support = ↓ Friend P3 amplitude
- ↑ support = ↓ Celebrity P3 amplitude
- ↑ support = ↓ Stranger P3 amplitude

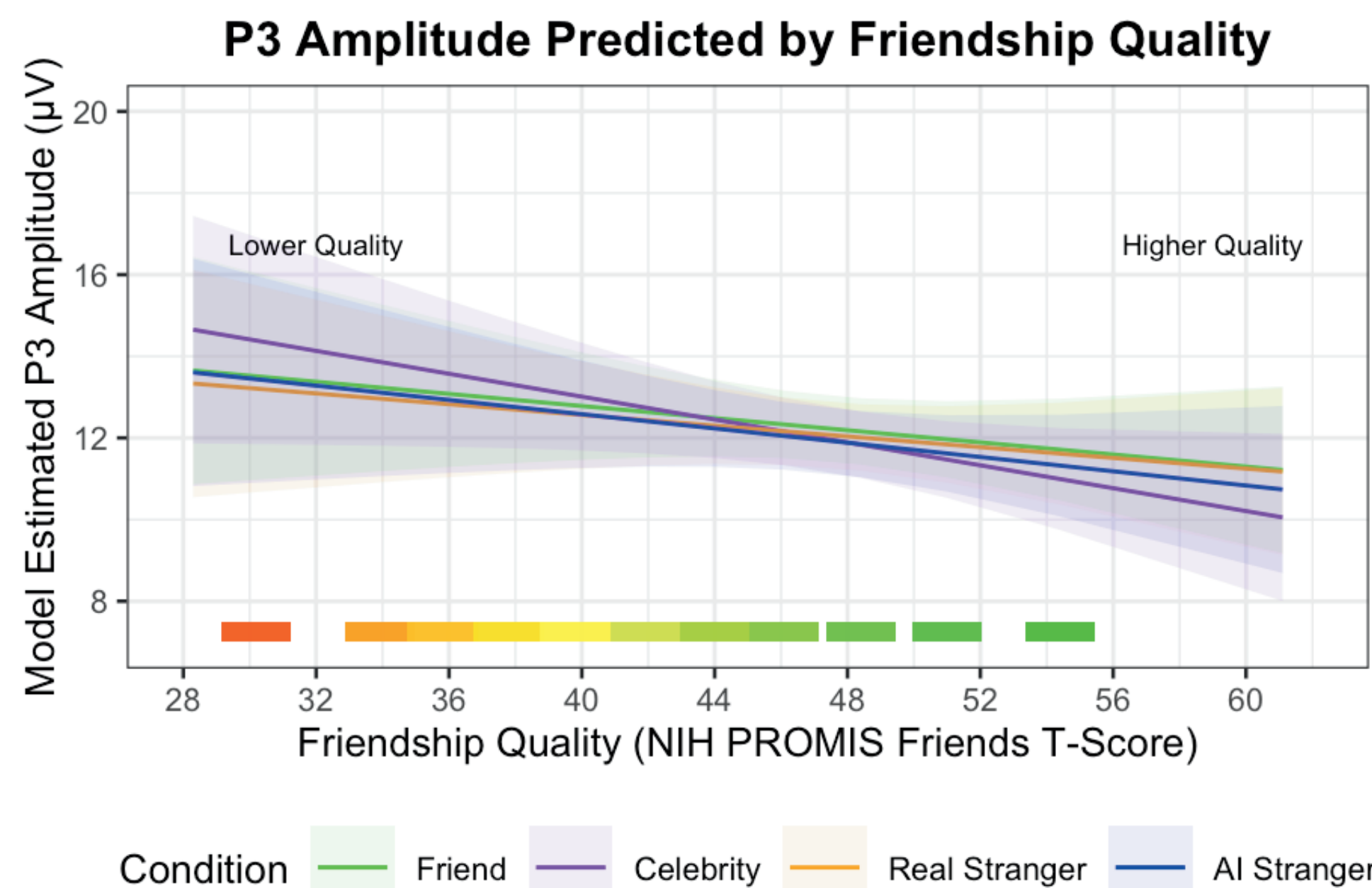


Increased friendship quality was linked to decreased cognitive processing: $F(1, 37) = 17.17$, $p < .001$

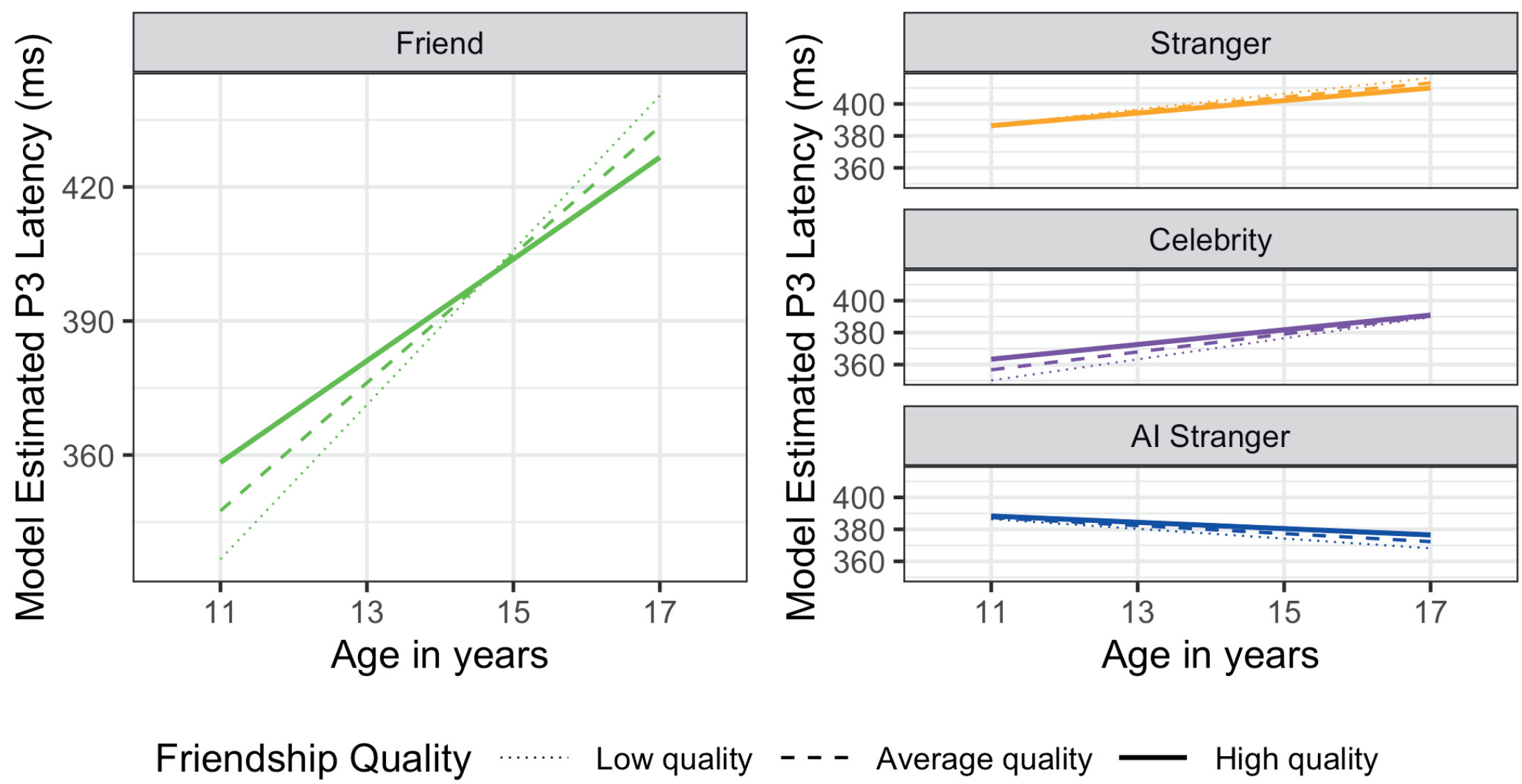
- ↑ Relationship quality ↓ P3 amplitude

The effect of friendship quality varied by condition, $F(1,35) > 3.86$, $p = .009$:

- Celebrity slope < Friend slope
- Celebrity slope < Stranger slope
- Celebrity slope < AI Stranger slope



Predicted P3 Latency by Friendship Quality



Among younger adolescents, higher friendship quality was associated with slower responses specifically to **friend faces** ($p = .002$), a pattern that was not observed in older adolescents or other conditions.



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MAIN TAKEAWAYS

Adolescents have extend or prolonged brain responses to their friend's face, which may suggest a **unique brain system for personally meaningful relationships**. This system is modulated by individual traits:

- Greater brain responses when teens report lower social support and friendship quality
 - Younger kids with lower friendship quality had more rapid responses to friends than older kids
- At lowest levels of friendship quality, we saw greater brain responses – specific to **celebrity faces**, which may suggest a compensatory mechanism

Next steps include evaluating how brain responses map on to daily experiences and other important social brain processes.